Task allocation sheet

DQN on Atari Breakout (ALE/Breakout-v5)

Member	Name	What they did	Files
Member 1	Celine Laura Ishimwe	GitHub repo and uv setup (pyproject.toml, uv.lock). make_env() with grayscale, 84x84, frame skip 4 and 4 stacked frames. Main training loop in train.py: networks, buffer, optimizer, target network update, CSV log, saving the model, command-line hyperparameters. Joining everyone's pieces.	train.py, pyproject.toml
Member 2	Ulrich Rukazambuga	QNetwork: 3 conv layers and 2 linear layers, input 4x84x84, one Q-value per action. play.py: loads dqn_model.pth and plays Breakout live. Apple GPU (MPS) support. Demo video of the best agent.	q_network.py, play.py
Member 3	Gentil Tonny Christian Iradukunda	ReplayBuffer (add, sample, __len__) with a deque. train_step: sample a batch, Q-values of the taken actions, Bellman target from the target network, Huber loss, backward and optimizer step. Test that the loss goes down on a fixed batch. This task allocation sheet.	replay_buffer.py, train_step.py, test_train_step.py
Member 4	MAHE Digne	epsilon_by_step and select_action (epsilon-greedy). plot_results.py for the best experiment. Results table template. README with the architecture diagram. check_repo.py and the final check. Best model copied to dqn_model.pth.	train.py (epsilon, actions), plot_results.py, check_repo.py, README.md

Experiments

Every member ran 10 experiments of 300,000 steps, changing one hyperparameter at a time (learning rate, gamma, batch size, epsilon start, end and decay). Each member's 10 models, 10 logs and results table are in their own folder: Laura/, Ulrich/, Tonny/ and Mahe/.